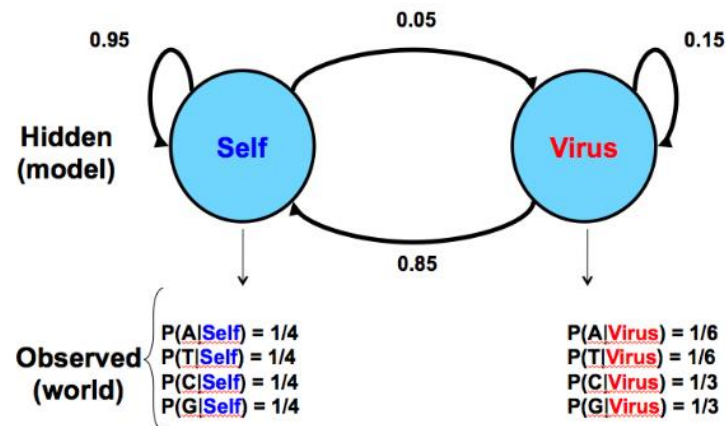


Exercise Sheet T10: Hidden Markov Models

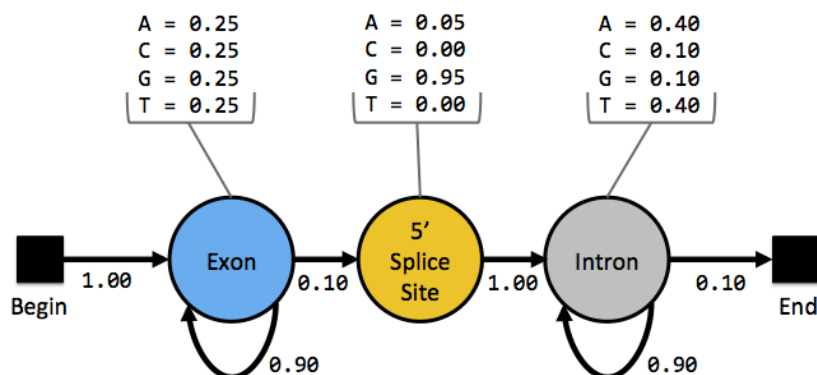
1. Considering this simple HMM model, answer the following questions?



- What are the hidden states in this HMM?
- What are the observable symbols (emissions)?
- What is the probability of transitioning from Self to Virus?
- Which state is more likely to generate longer sequences? Why?
- Write down the *transition* and *emission* matrices for this model.
- Given the path S – S – V – V and observed sequence A C G G, compute:
 - $P(\pi)$
 - $P(x|\pi)$
 - $P(x, \pi)$
- You find a stretch of DNA with many C's and G's. What does this model suggest about its origin?
- Write a simulator to generate sequences from this HMM. For each simulated sequence, track the hidden state and the observable emission. Create a 10000-nucleotide sequence and compare the observed CG content with your expectations. Do Virus states consistently emit more C and G than Self? Plot the CG content across the sequence (*you can use a sliding window of 50bp*). Can you visually detect transitions between states?

The following exercise should be submitted on Moodle.

2. Consider the following HMM model:



- a. Because an HMM is based on probabilities, there are many possible sets of paths that could have generated the same nucleotide sequence. Based on the schema below and the HMM diagram provided, **determine which of the splice sites is the most likely**. How confident are you in the *best path*?

